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A Late Quaternary Geomagnetic Secular Variation Record From Lake Waiau, Hawaii, and the Question of the Pacific Nondipole Low

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A continuous paleomagnetic record for the last 13,000 years was obtained from the sediments of Lake Waiau, located near the summit of Mauna Kea Volcano on the island of Hawaii. The secular variation (SV) of the geomagnetic field was studied using spectral analysis and precession analysis of the magnetic vector. The hypothesis that there has been anomalously low geomagnetic SV in the central Pacific was tested using angular dispersion analysis. The inclination record has a mean value of 27° that is much lower than the 35° predicted by the geocentric axial dipole hypothesis. The spectrum obtained from spectral analysis shows major inclination peaks at periodicities of approximately 3600 and 7900 years and major declination peaks at periodicities of approximately 1100, 2000, 3300, and 7800 years. The approximately 8000- and 3500-year periodicities shown in both inclination and declination spectrums may represent the dipole and nondipole variation, respectively. The approximately 1100- and 2000-year periodicities are probably derived from either the sources that affect declination only or the offsets between the core sections. In addition, Bauer plots of the magnetic vector are dominated by clockwise motion. The angular dispersion calculated from lake sediments is 5.9° with respect to the mean virtual geomagnetic pole (VGP) and 9.1° with respect to the geographic pole. Both values are much lower than the 13.5° expected from paleosecular variation model G which separates the angular dispersion into components from the dipole and the quadruple families. Our results support the hypothesis that an anomalously low secular variation is a characteristic feature of the central Pacific. Furthermore, this geomagnetic behavior persists on time scales of 10^4 years. Regional anomalies in temperature, topography or electrical conductivity at the core-mantle boundary are inferred to cause the low SV.

INTRODUCTION

Secular Variation and the Objectives of the Study

The study of geomagnetic secular variation (SV) can aid in our understanding of the Earth's interior and the origin of the Earth's magnetic field. A measure of SV is either the angular standard deviations (angular dispersion) of paleofield directions or their associated virtual geomagnetic poles (VGPs) [*Cox and Doell*, 1960]. The VGP method is used in this study because it allows an easier separation of the effects of dipole wobble and nondipole field sources, and the angular dispersion of VGPs due to dipole wobble is invariant with respect to latitude [*Cox*, 1962; *Cox and Doell*, 1964].

The major objectives of this study are (1) to test the hypothesis that subdued SV and a subdued nondipole field exist in the central Pacific on time scales of 10^4 years by angular dispersion analysis of the continuous, radiocarbon-dated paleomag-netic record obtained from the sediments of Lake Waiau, (2) to identify the periodic components of secular variation by spectral analysis of the inclination and declination records, and (3) to examine the looping behavior of the geomagnetic vectors to infer the nature of secular variation.

The Pacific Nondipole Low

The SV of the present-day geomagnetic field is lower in the Pacific Ocean region than in the remaining part of the world [e.g., *Fisk*, 1931; *Fleming*, 1939; *Chapman and Bartels*, 1940; *Runcorn*, 1956; *Depietri*, 1961; *Vestine and Kahle*, 1966; *Doell and Cox*, 1971; *Bingham and Stone*, 1972]. The results of early paleomagnetic studies on Brunhes age Hawaiian lava flows (<0.7 Ma) were characterized by low angular dispersion of the VGP. These estimates of SV led *Doell* [1969, 1972*a*, *b*] and *Doell and Cox* [1961, 1963, 1965, 1971, 1972] to propose that a subdued nondipole field exists in the Pacific region. Studies conducted elsewhere in the Pacific, including the Galapagos Islands [*Doell and Cox*, 1972], Easter Island [*Isaacson and Heinrichs*, 1976], Society Island [*Duncan*, 1975], revealed that SV is consistent with a "standard" mode of global variation based on modern dipole and nondipole fields [e.g., *Cox*, 1970]. These results constrained the region of anomalously low secular variation to a correspondingly smaller region of the Pacific [*McWilliams et al.*, 1982].

Different interpretations of the paleomagnetic data from lava flows arise because the eruption of lava flows is episodic, resulting in a lower resolution SV curve. Furthermore, sufficient stratigraphic control was not always attainable in the early studies of lava flows, resulting in imprecision of the dating. These considerations and more recent data from lava flows led *Duncan* [1975], *McElhinny and Merrill* [1975] and *Coe et al.* [1978] to conclude that the low nondipole field in Hawaii is either a recent phenomenon, probably restricted to Holocene time, or an artifact of incomplete sampling. *McWilliams et al.* [1982] studied ^{14}C dated lava flows on Hawaii. The angular dispersion they calculated:

$$11.2^{+0.9}_{-0.8}{}^{\circ}$$

is significantly lower than the value of 13.5° predicted by a recent global SV model (model G) that describes the relationship between VGP angular dispersion and latitude [*McFadden et al.*, 1988]. Despite improvements in dating, the distribution of ^{14}C dated data [*McWilliams et al.*, 1982] is strongly biased toward younger lava flows. In addition, the nonuniform temporal distribution of the data affects the calculation of the angular dispersion. Fortunately, both the dating problems and the nonuniform temporal distribution of the data encountered in lava flow studies can be overcome in lake sediment studies.

The Geomagnetic Spectrum

Denham [1975] first suggested that the quasi-periodic fluctuations in the geomagnetic field recorded in lake sediments

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could be analyzed by power spectrum techniques. Problems associated with the time series analysis of paleomagnetic data from lake sediments include (1) dating errors, (2) changes in sedimentation rate between dated intervals, and (3) an insufficient core length available to derive low frequency components of the power spectrum. *Barton* [1982] made a preliminary synthesis of spectra for observatory, archaeomagnetic, lake sediment, and marine sediment records and concluded that no distinct boundary exists between the inferred frequency domains of dipole and nondipole field effects and the spectrum has a broad maximum between 10^4 and 10^5 years.

Drift Direction of the Geomagnetic Field Inferred From the Paleomagnetic Data

Directions of the geomagnetic field are often represented on Bauer plots of inclination versus declination or equivalent VGP circularity. Bauer plots and VGP circularity are sometimes used to infer the drift direction of the geomagnetic field [*Runcorn*, 1959; *Skiles*, 1970].

Runcorn [1959] showed that a moving magnetic source in the Earth's core can cause the tip of the magnetic vector at a fixed observatory on the Earth's surface to rotate clockwise (when viewed from tail to tip of the vector) for a westward motion of the source and counterclockwise for an eastward motion of the source. This phenomenon is known as Runcorn's rule. *Skiles* [1970] extended the argument to radial dipole source models of the nondipole field and concluded that exceptions to Runcorn's rule are rare, if they do exist. *Dodson* [1979] has drawn attention to the nonuniqueness of the clockwise/westward drift interpretation, but Runcorn's rule is still applicable for a wide range of field configurations [*Barton*, 1983].

GEOLOGICAL SETTING

Lake Waiau is a volcanic crater lake situated in Puu Waiau (19°48'N, 155°29'W) at an altitude of 3970 m on Mauna Kea, Hawaii (Figure 1). It has a surface area of 0.73 ha and a water depth of about 3 m [*Maciolek*, 1969]. The area surrounding Lake Waiau is a desert with sparse herbaceous vegetation and poorly developed soils. The lake flora consists of a few species of planktonic diatoms and algae and an epilithic mat of primarily amorphous blue-green algae [*Maciolek*, 1969; *Massey*, 1978]. The fauna consists of one species of cladocera and two benthic species, a worm and a fly larvae [*Maciolek*, 1969]. The benthic species are abundant and graze on the epilithic algae mat but probably do not bioturbate the underlying sediments. The sediments from Lake Waiau are distinctly laminated and composed mainly of lacustrine silt, clay, and volcanic ash. The persistence of fine (1-2 mm thick) laminations throughout the sediment column indicates that bioturbation is minimal in Lake Waiau.

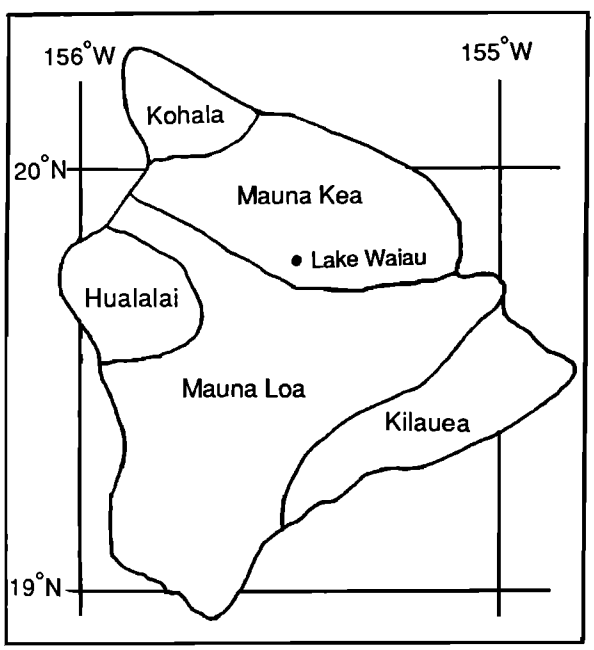


Fig. 1. Location map of Lake Waiau on the island of Hawaii.

METHODS AND RESULTS

Coring and Magnetic Measurements

Three cores (82-1, 82-2, 82-3) were recovered from Lake Waiau using a modified Livingston piston corer. The core orientation was obtained by using locking extension rods and a rectangular top extension rod that cannot twist relative to the corer. The orientation of the top rectangular rod was measured in each core. Cores 82-1 and 82-3 suffer from orientation errors caused by mechanical problems with the locking extension rods that allowed sporadic twisting in the horizontal plane. On the other hand, core 82-2 was properly oriented throughout and therefore should have a high-quality declination record.

Two replicate 5 cm^3 box samples were taken continuously at a 2-cm interval down each core. The samples were measured using either a Schonstedt spinner magnetometer or a SCT cryogenic magnetometer before and after demagnetization in an alternating-field (AF) with a peak value of 10 mT. Typical Zijderveld plots (Figure 2) indicate that the natural remnant magnetization (NRM) of Waiau sediment is single-component and stable. The Zijderveld plots also indicate that secondary components have typically been removed by AF demagnetization in a peak field of 10 mT. For this reason, a blanket AF treatment of 10 mT was applied to all samples.

Isothermal remnant magnetization (IRM) acquisition curves for Lake Waiau sediments indicate that the dominant magnetic carrier is magnetite or a similar composition (Figure 3). Curie temperatures of magnetic seperates from Lake waiau sediments range between 518° and 533°C [*King et al.*, 1982] and indicate that the magnetic carrier is probably titanomagnetite. In addition, the results of *King et al.* [1982] indicate that the average magnetic particle size of the lacustrine silt and ash/silt layers is in the psuedo-single domain (PSD) range and does not vary significantly with depth. However, the volcanic ash layers are significantly coarser and contain magnetic material in the multidomain size range. Prior to further analysis, replicate samples from coarse ash layers and from horizons that had a difference of the two replicate samples greater than 10° (about 10% of the total) were eliminated to reduce the possible effects of core disturbance and poor recording of the geomagnetic field. Figures 4 and 5 show inclination and declination records for the remaining data from the three cores versus depth, respectively.

Age Control

The sediments of Lake Waiau are ideal for accurate radiocarbon dating because of the relative absence of either old carbon or carbonate sources in the watershed. Age control for Lake Waiau sediments was determined by five ^{14}C dates from bulk samples from core 82-1. We selected core 82-2 as the reference core for stacking the paleomagnetic data because of its superior azimuthal orientation. Therefore the ^{14}C dates need to be converted to the equivalent depths in core 82-2. A high-resolution depth relationship between core 82-1 and core 82-2 was readily established using the very consistent lamination pattern. The corresponding depth in

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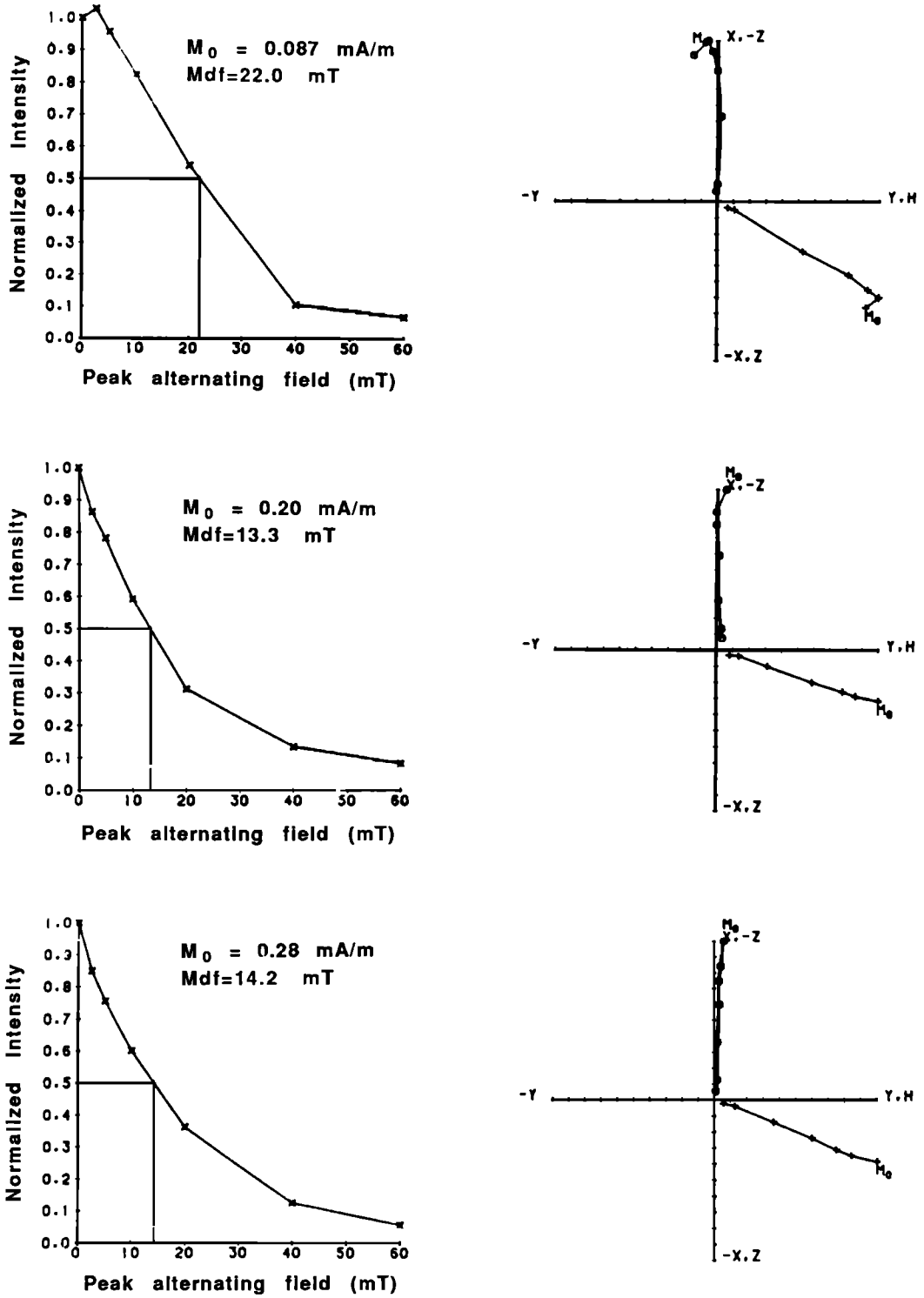


Fig. 2. Typical Zijderveld plots for samples from Lake Waiau core 82-2 and their demagnetization curves. These plots indicate that the natural remanent magnetization (NRM) of Waiau sediment is stable and single-component.

core 82-2 to the depth in the radiocarbon-dated core 82-1 was calculated from the linear regression equation of the depth of laminae in core 82-2 to the depth of the same laminae in core 82-1 (Figure 6). The five ^{14}C dates from core 82-1 and their equivalent depths in core 82-2 are listed in Table 1. Two age models, a linear regression model and a linear interpolation model, were formulated (Figure 7).

Construction of a Composite Secular Variation Curve

Two approaches were used to construct a composite curve of geomagnetic SV from Lake Waiau. In each approach, a reference core is chosen and the depth scales of the other cores (called subsidiary cores) are adjusted relative to the reference core. The first approach is to transform the depth scales of the subsidiary cores to

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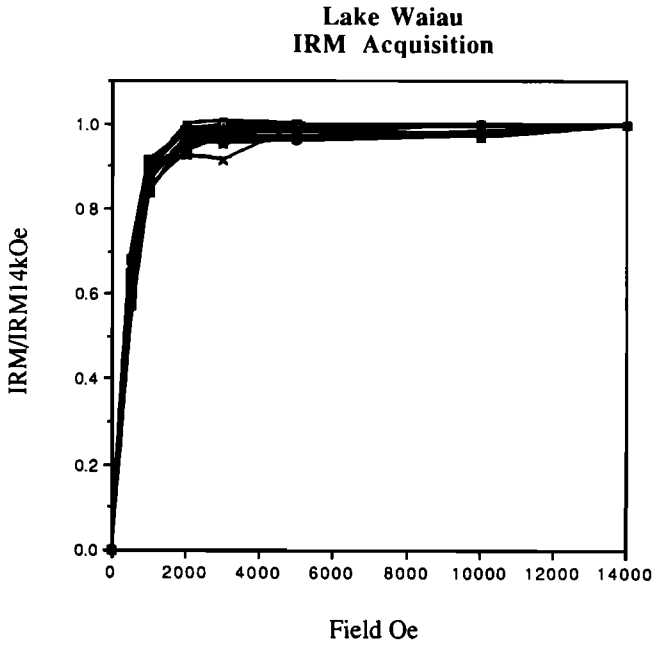


Fig. 3. Isothermal remanent magnetization (IRM) acquisition curves for Lake Waiau sediments. These curves indicate that the dominant magnetic carrier is magnetite or a similar composition.

the equivalent depth in the reference core using the depth relationship determined by the pattern of laminations. The inclination and declination data from the subsidiary cores were then linearly interpolated to the depth scale of the reference core. An average value of inclination and declination from the reference and subsidiary cores at each equivalent horizon was then calculated. The second approach is to use CORPAC, a computer program designed for signal correlation [*Martinson et al.*, 1982, 1987]. Using CORPAC, the inclination records from the three Waiau cores were visually correlated (Figure 8). In addition, the declination records were independently correlated (Figure 9). The mapping functions and correlation coefficients derived from both the inclination and declination correlations are shown in Figure 10. The mapping functions of the inclination and declination correlations are fairly similar. However, the pattern of the declination records is somewhat inconsistent because of orientation errors in cores 82-1 and 82-3, and consequently, the correlation coefficient is low.

The stacked inclination and declination curves and their standard deviations using linear interpolation after lithologic correlations are made and CORPAC are shown in Figures 11 and 12. The general shapes of the stacked data using the two methods are very similar. From the standard deviation comparison, it appears that the stacked inclination using linear interpolation of core laminations is slightly superior to the stacked inclination using CORPAC. The declination records from cores 82-1 and 82-3 are not very reproducible to that from core 82-2, especially for the data below 4

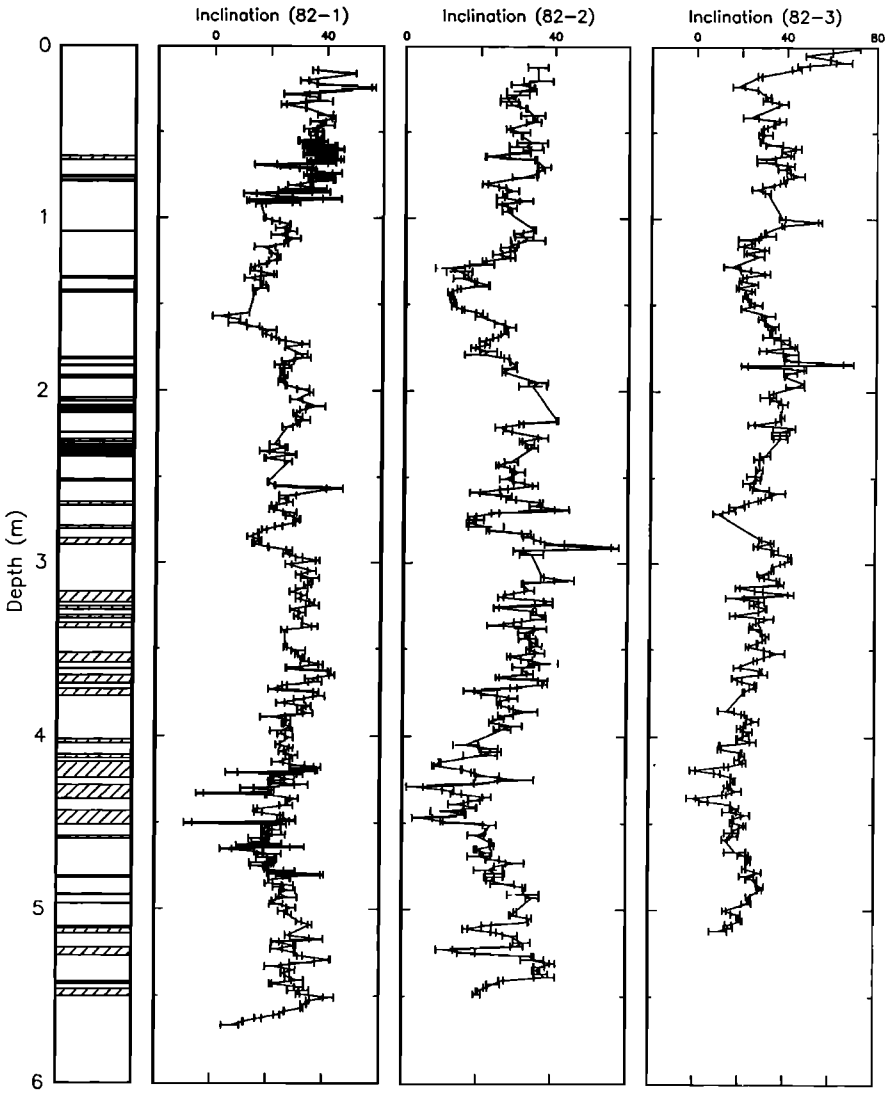


Fig. 4. Inclination versus depth from core 82-1, 82-2 and 82-3. The two replicate samples at each horizon are expressed using error bars. The lithologic stratigraphy of core 82-2 is shown on the left. Solid area represents volcanic ash, open area represents lacustrine silt, and hatchured area represents ash and silt.

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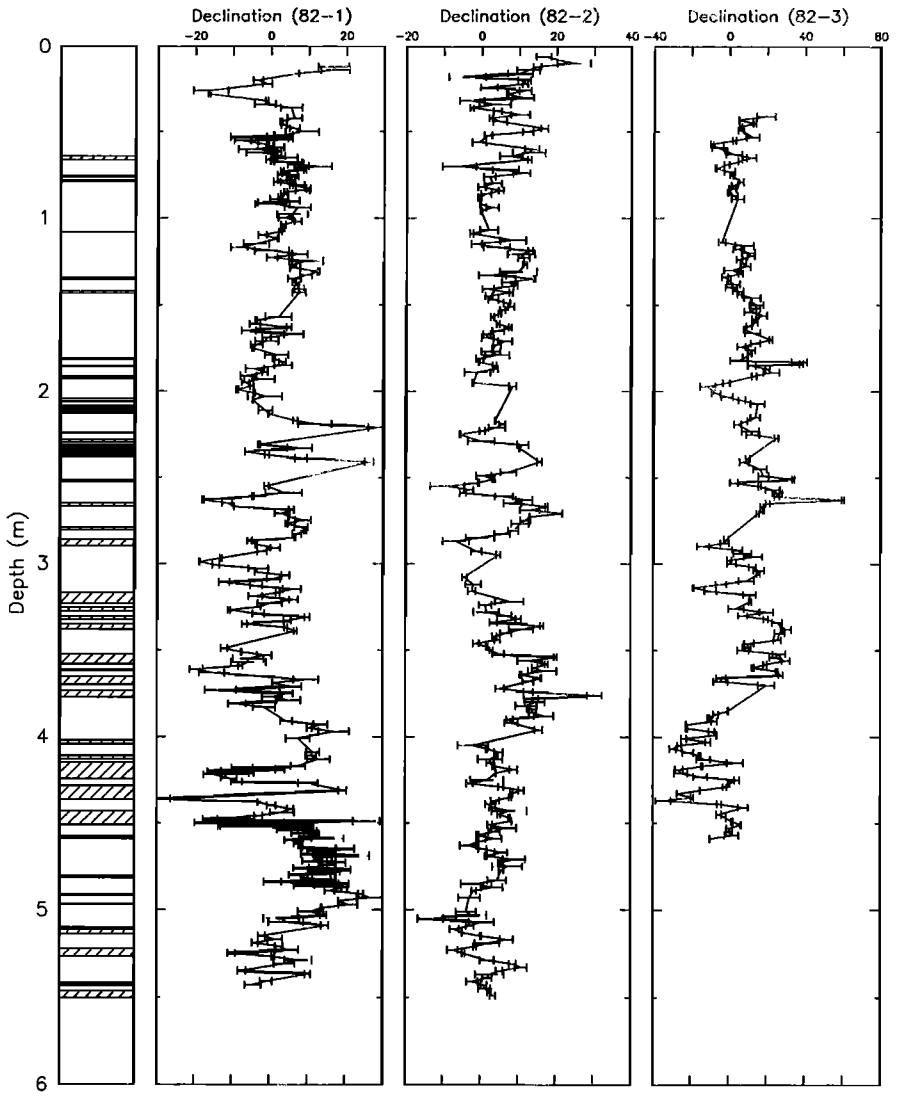


Fig. 5. Declination versus depth from core 82-1, 82-2, and 82-3. See Figure 4 caption for complete description.

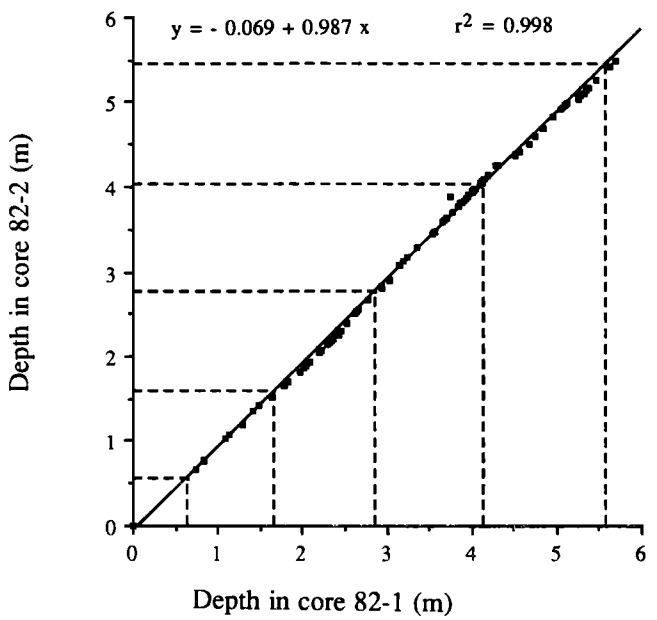


Fig. 6. Linear regression of the depths of correlated laminations in core 82-2 on their equivalent depths in core 82-1.

m in the section (Figure 9) due to core orientation problems. In this case, the stacked declination record from all three cores is probably inferior to the declination record from core 82-2. For these reasons, we use the stacked inclination record constructed by the lithologic correlation and linear interpolation approach and the declination record from core 82-2 as the best estimates of the paleosecular variation (PSV) curves for Hawaii.

Figure 13 shows the stacked inclination curve and the declination from core 82-2 after converting the depth scale to an age scale using

TABLE 1. Radiocarbon Dates Versus Depth in Core 82-1

Depth, m		
Core 82-1	Core 82-2	Age, years B.P.
0.64±0.05	0.56±0.05	400±160
1.67±0.05	1.58±0.05	3,200±180
2.88±0.05	2.77±0.05	7,160±180
4.14±.0.05	4.02±0.05	9,410±380
5.59±0.06	5.45±0.06	12,660±460

The corresponding depths in core 82-2 were determined from the linear regression equation of the depths of correlated laminations in core 82-2 to their equivalent depths in core 82-1.

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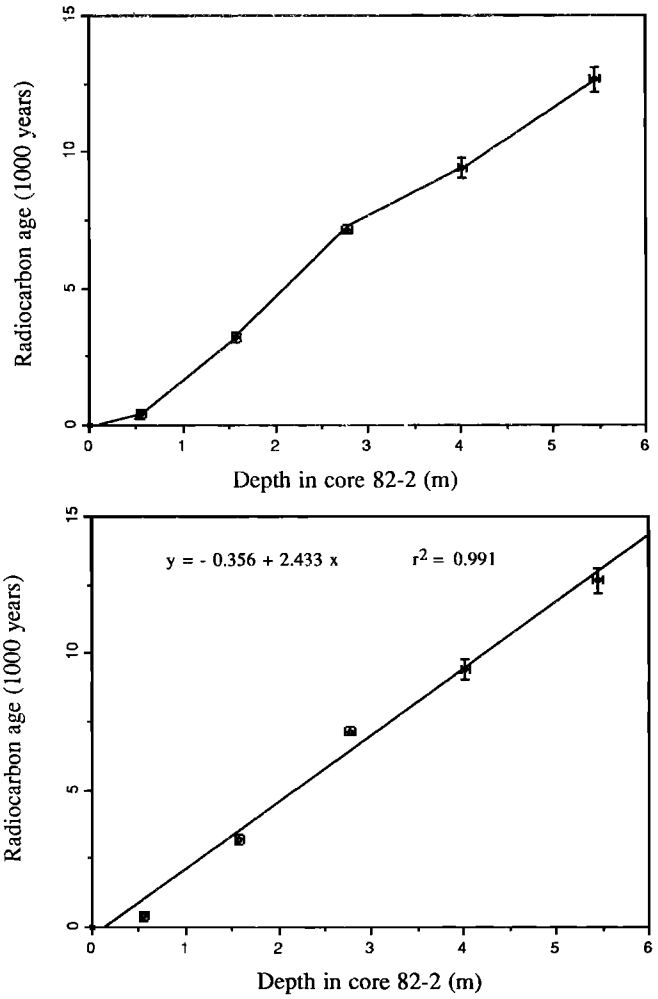


Fig. 7. Linear interpolation and linear regression age models for core 82-2.

the linear interpolation and linear regression age models, respectively. We favor the linear interpolation age model based on the comparison with the radiocarbon-dated lava flow data (see discussion). Figure 14 shows the PSV curve for Hawaii, that is, the stacked inclination curve and the declination curve from core 82-2 after transforming depth to age using the linear interpolation age model. The pattern of the Hawaii PSV curve shows three repetitive waveforms α, β, λ each with an interval of about 3500 years. This pattern may suggest a stable wave dynamo process. The standard paleomagnetic parameters of field direction and VGP calculated using the original data and the data shown in Figure 14 are listed in Tables 2 and 3, respectively. The VGP projections shown in Figure 15 indicate a "far-sided, right-handed" tendency [e.g., *McWilliams et al.*, 1982] with a mean value that departs from the geographic pole by 6.9°.

Angular Dispersion Analysis

The angular dispersion was calculated both by using a Fisherian precision parameter and by using an equation that does not assume a spherical normal probability distribution [*Cox*, 1970]. For comparison, the angular dispersion was calculated relative to both the mean VGP and the geographic pole. Fisherian statistics and the statistical approach that does not assume a normal probability distribution gave essentially the same results. Before doing the calculations, the data were thinned by using three averaging windows: 100 years, 250 years, and 500 years. The reasons for windowing the data are (1) to obtain equally spaced data and (2) to lessen the effect of spurious data by using averaged VGPs. Table 4 lists the results of the angular dispersion calculations. The 95% confidence limits for the angular dispersion are determined from *Cox* [1969]. The angular dispersion with respect to the mean VGP is 5.9°. This value is much smaller than the 9.1° calculated with respect to the geographic pole probably because of the biased distribution of the VGPs relative to the geographic pole. However, no matter how the angular dispersion is calculated, it is always significantly lower than the 13.5° calculated for the same latitude range from a world-wide data set for the past 5 m.y. [*McFadden et al.*, 1988]. The continuous paleomagnetic data from lake sediments

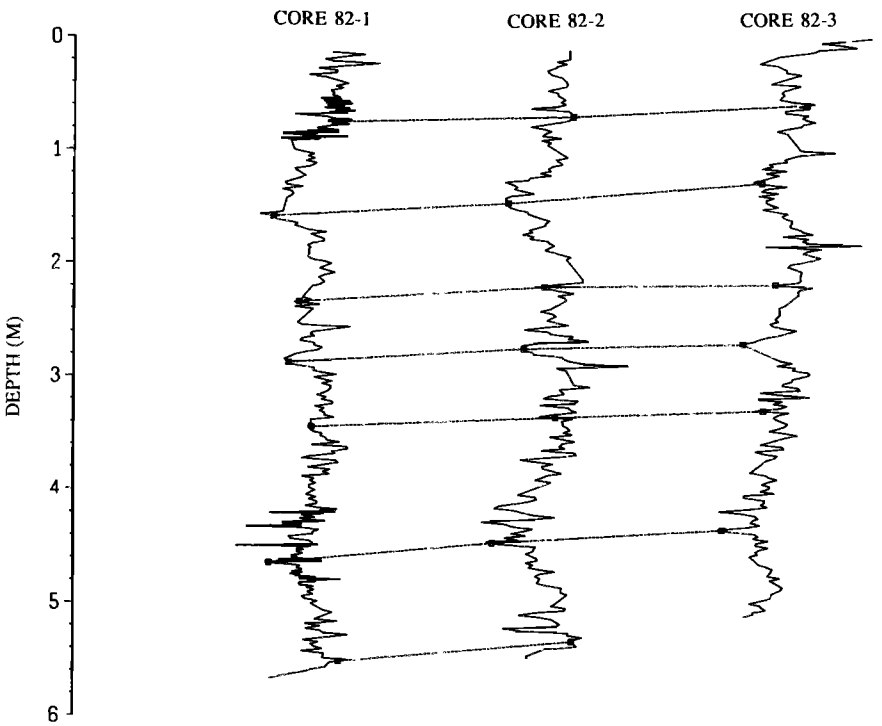


Fig. 8. Inclination correlations between cores.

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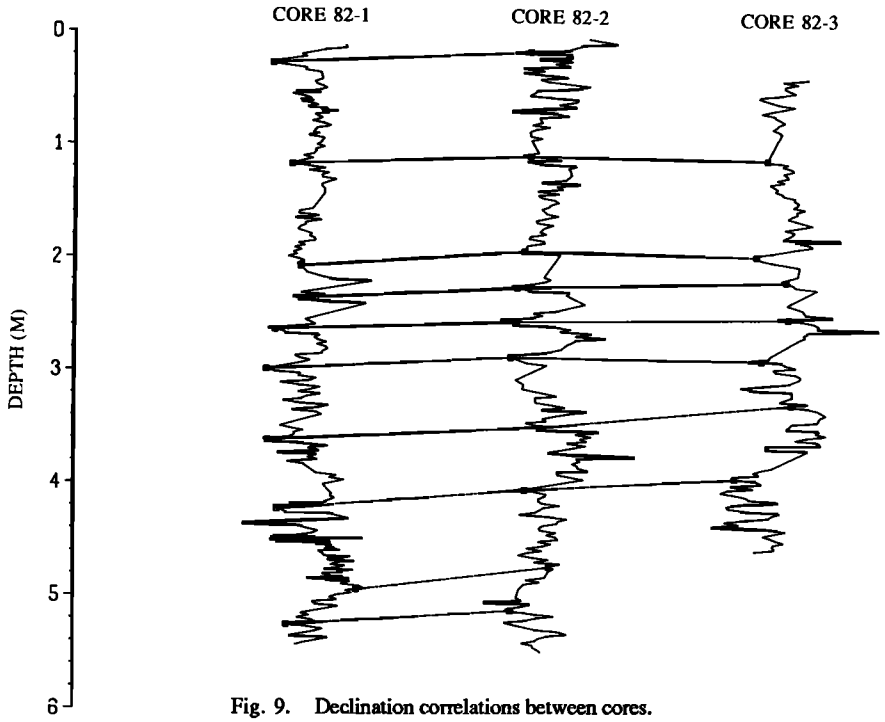
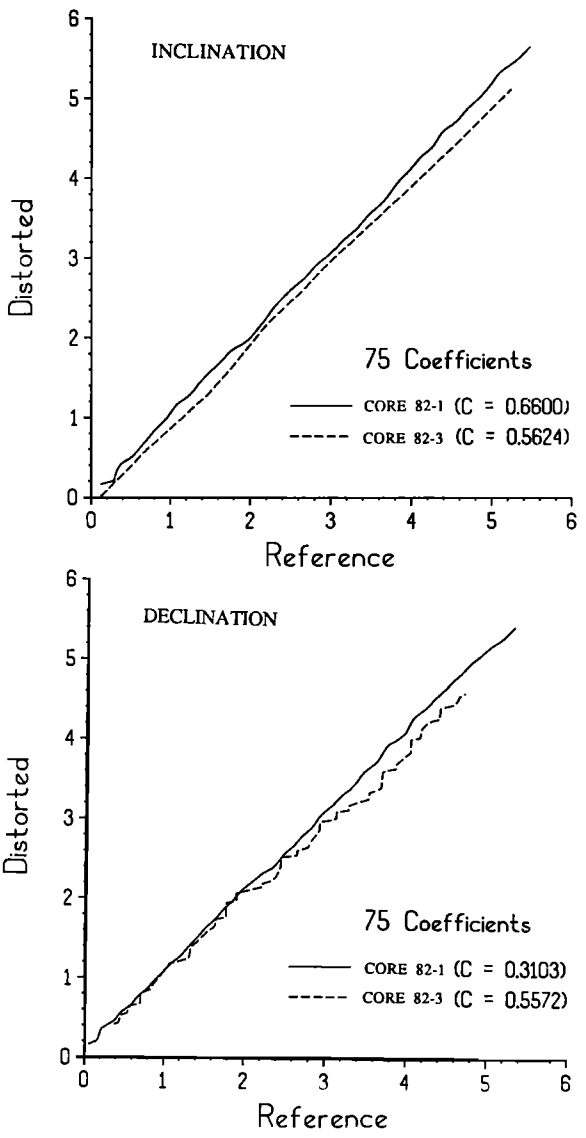


Fig. 9. Declination correlations between cores.



show anomalously low angular dispersion in Hawaii for the past 13,000 years, as does the lava flow data. Furthermore, the angular dispersion calculated from the lake sediment data is lower than the 11.2° with respect to the mean VGP calculated from the lava flow data [*McWilliams et al.*, 1982].

Scalar Spectral Analysis

Spectral analyses of inclination and declination data from core 82-2 were performed using both the periodogram method described by *Bloomfield* [1976] and Burg's maximum entropy method (MEM) [*Burg*, 1967, 1968]. Both the periodogram method and the MEM have their advantages and disadvantages. The periodogram method does not require subjective choices of the parameters, but it requires strong assumptions about the nature of the time series outside the observed range, and this can lead to poor resolution and peak shifting in short series [*Toman*, 1965]. The MEM does not require us to extend the time series, but the resulting spectrum is often sensitive to the largely subjective choice of the parameter m, which is the order of autoregressive process used to model the data [*Barton and McElhinny*, 1982]. Therefore, we decided to use both methods to perform spectral analyses of inclination and declination time series.

Prior to the analyses, the data were linearly interpolated to a 60-year interval. When the periodogram method was used, the time series was cosine tapered 10% at each end before Fourier transformation in order to minimize the effects of data truncation. When the MEM was used, we chose *Ulrych and Clayton's* [1976] criterion: $m_{UC}=N/3$ to $N/2$ for the present work, where N is the length of the data set.

The spectrums obtained using both the periodogram and the MEM are shown in Figures 16 and 17. The frequencies and their equivalent periodicities of major peaks are listed in Table 5. The frequencies shown in Table 5 are the averaged frequencies of the peaks obtained from the periodogram method and MEM. The major inclination peaks have periodicities of approximately 3600 and 7900 years, and the major declination peaks have periodicities of approximately 1100, 2000, 3300, and 7800 years.

Fig. 10. Mapping functions of depth from inclination and declination correlations using CORPAC.

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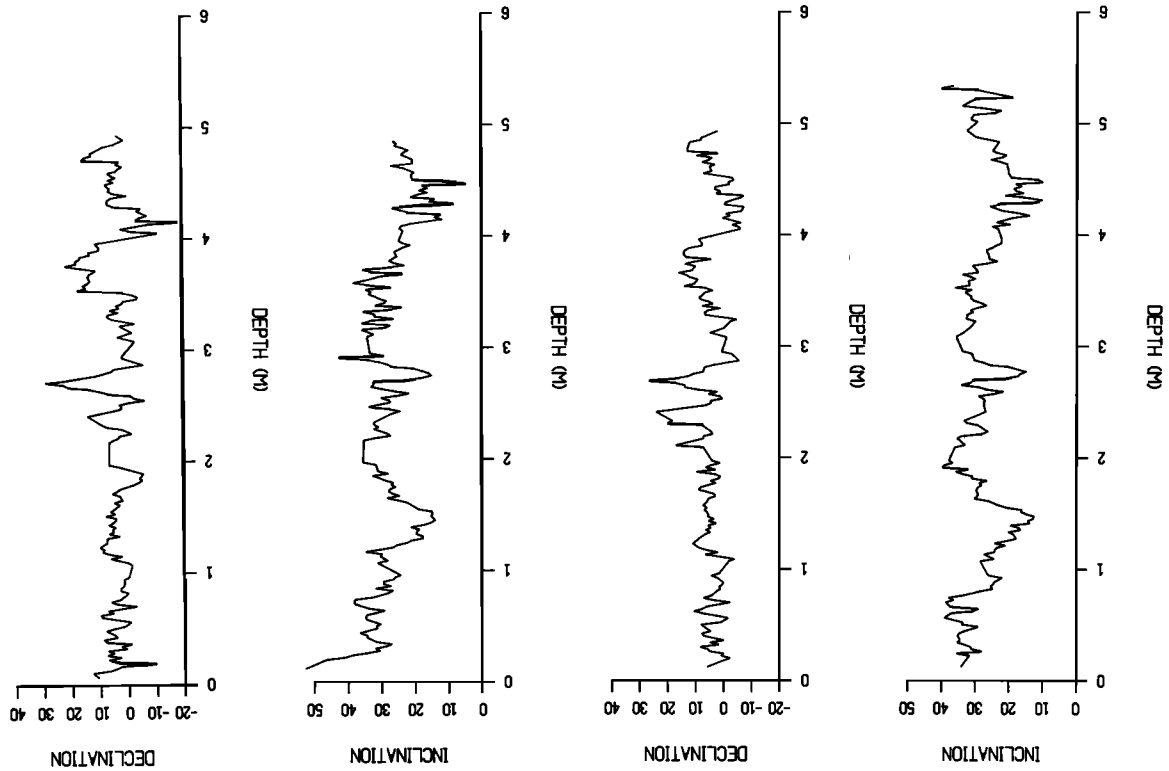


Fig. 11. Stacked inclination and declination using linear interpolation (left two) and CORPAC (right two).

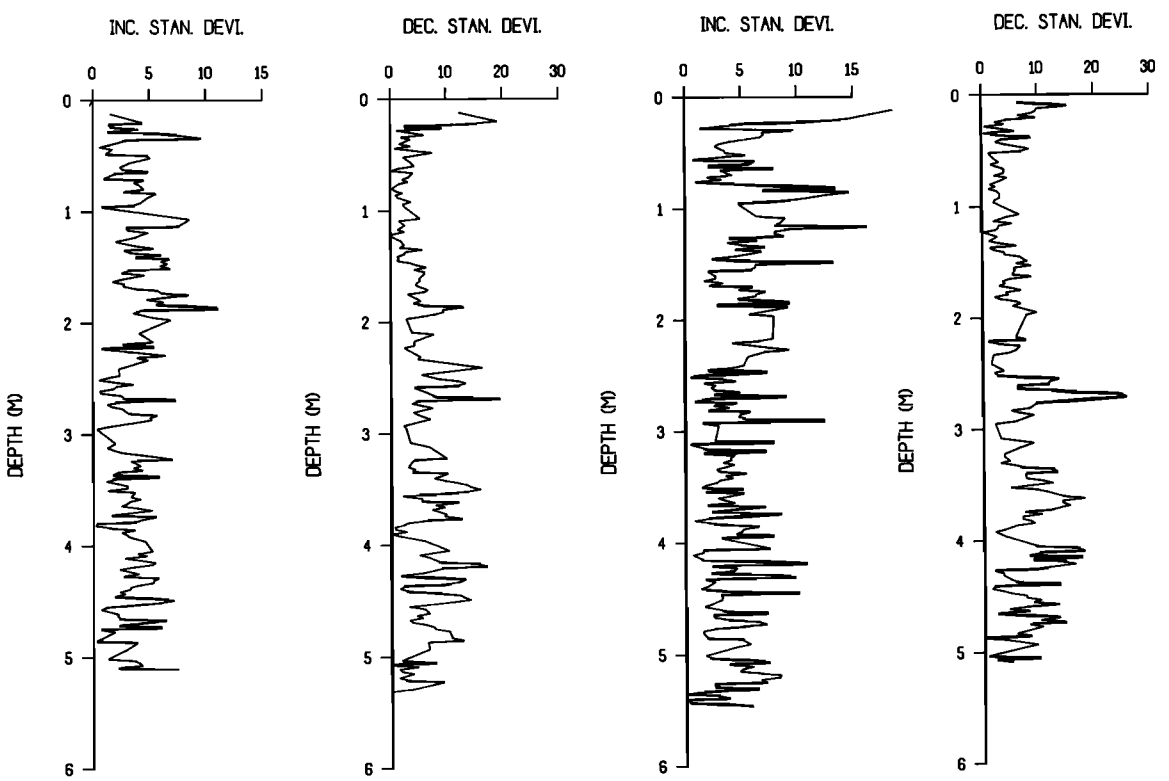


Fig. 12. Stacked inclination and declination standard deviations using linear interpolation (left two) and CORPAC (right two).

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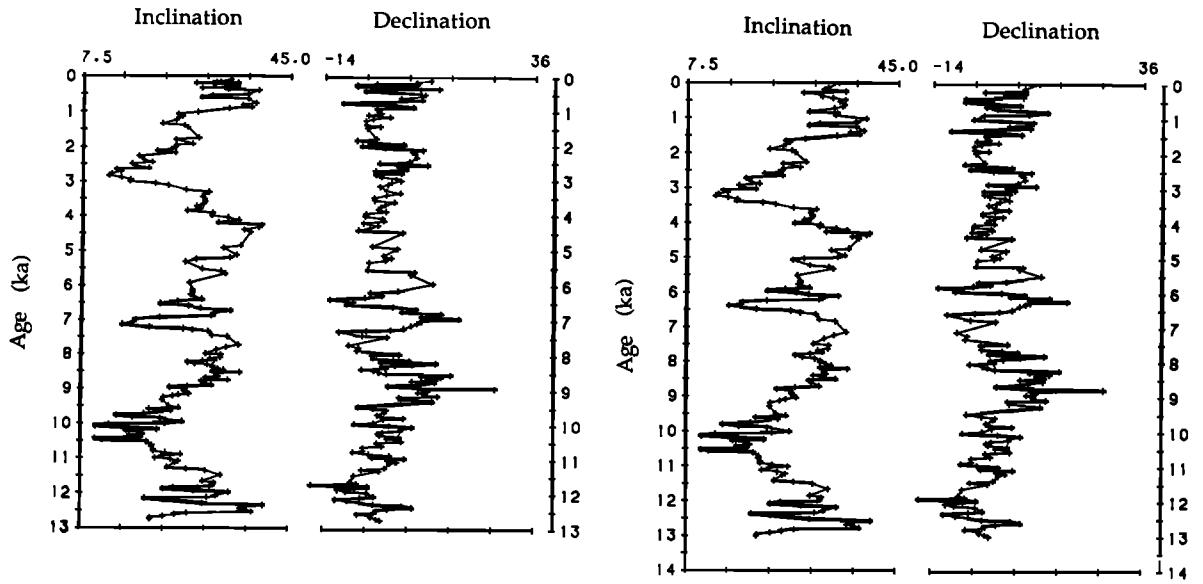


Fig. 13. Inclination and declination versus age derived from linear interpolation age model (left) and linear regression age model (right).

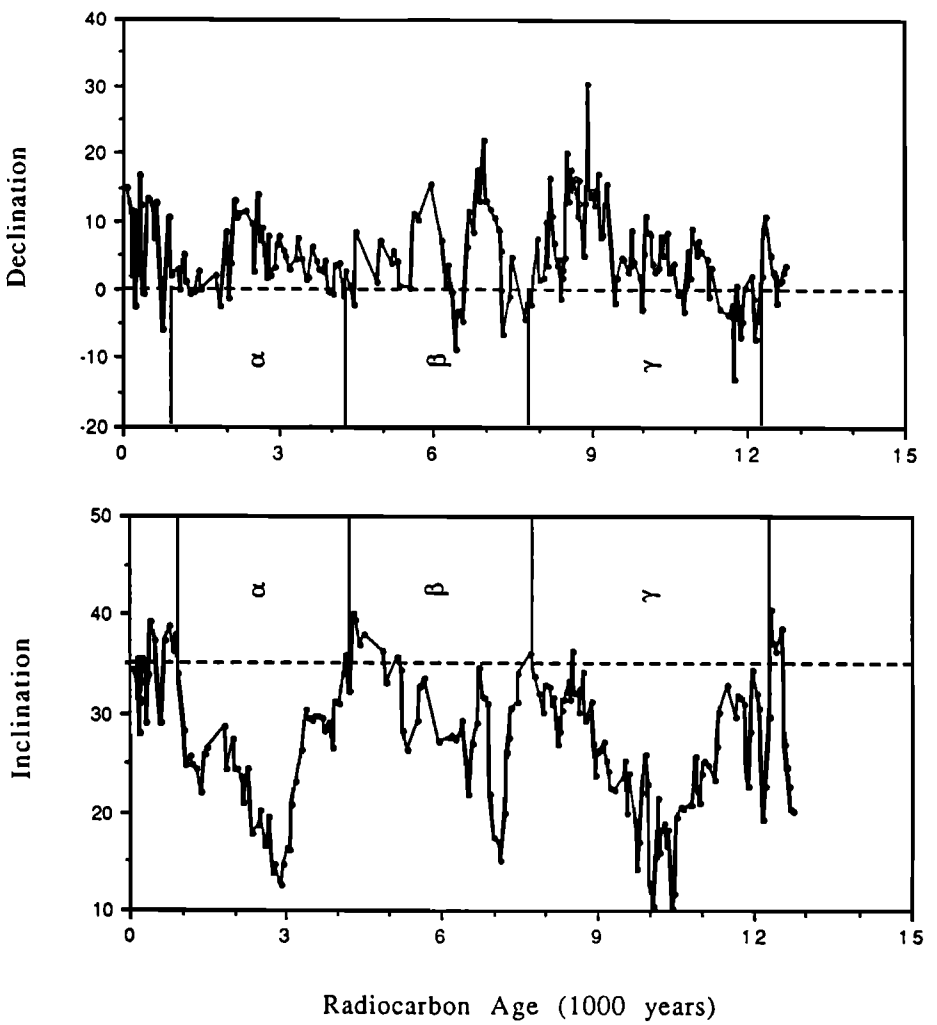


Fig. 14. Paleosecular variation curve for Hawaii. Note the periodic character of inclination and declination.

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TABLE 2. Standard Paleomagnetic Parameters

	Core 82-1			Core 82-2			Core 82-3		
	100-Year Window	250-Year Window	500-Year Window	100-Year Window	250-Year Window	500-Year Window	100-Year Window	250-Year Window	500-Year Window
Number of data	114	49	25	112	50	26	90	41	22
Average inclination	26.9	26.4	26.3	27.3	27.4	28.0	30.0	30.1	29.6
Average declination	2.9	2.8	2.8	4.6	4.8	4.7	5.1	7.1	6.1
Parameters of field directions									
k	67.6	88.4	93.3	81.2	100.2	133.8	23.2	26.5	35.1
s	9.9	8.6	8.4	9.0	8.1	7.0	16.8	15.7	13.7
α_{95}	1.6	2.2	3.0	1.5	2.0	2.5	3.2	4.4	5.3
Position of mean VGP									
λ	83.9	83.7	83.6	83.3	83.2	83.5	83.9	82.2	82.8
Φ	-3.6	-1.4	-1.5	-18.1	-19.4	-20.1	-34.0	-42.5	-36.8
Parameters of VGP									
K	82.2	119.4	122.7	132.8	168.0	236.0	24.8	27.4	40.1
S	8.9	7.4	7.3	7.0	6.3	5.3	16.3	15.5	12.8
α_{95}	1.5	1.9	2.6	1.2	1.6	1.8	3.1	4.3	5.0

Calculations are done for each core after thining the data with three windows.

Precession Analysis of the Geomagnetic Vectors

Bauer plots of inclination versus declination were used to examine the looping behavior of the geomagnetic vectors and to infer the drift direction of the geomagnetic field. Bauer plots are shown in Figure 18. From approximately 12,700 years B.P. to 11,300 years B.P., the field vectors show counterclockwise precession. The generalized precession pattern of the field vectors is shown on Figure 19. From approximately 11,300 years B.P. to 700 years B.P., clockwise precessions are evident. The precession pattern is not clear for the last 700 years.

TABLE 3. Standard Paleomagnetic Parameters Calculated From the Stacked Inclination Record and the Declination Record From Core 82-2 Using Three Window Sizes to Thin the Data

	Size of Window, years		
	100	250	500
Number of data	112	50	26
Average inclination	27.2	27.3	27.6
Average declination	4.7	4.9	4.7
Parameters of field directions			
k	97.7	116.3	153.9
s	8.2	7.5	6.5
α_{95}	1.4	1.9	2.3
Position of mean VGP			
λ	83.1	83.0	83.3
Φ	-17.3	-18.8	-18.6
Parameters of VGP			
K	148.0	183.2	255.2
S	6.7	6.0	5.1
α_{95}	1.1	1.5	1.8

DISCUSSION

Selection of the Age Model

The most recent paleomagnetic data from lava flows in Hawaii [*Holcomb et al.*, 1986] show an inclination peak at about 1800 years B.P. (Figure 20). From Figure 13, the same peak in the lake sediments is younger than 2000 years when the linear interpolation age model is used and older than 2000 years when the linear regression age model is used. Similarly, an inclination trough at about 2300 years B.P. in the lava flow record has an age younger than 3000 years from the linear interpolation age model and older than 3000 years from the linear regression age model. Therefore, we prefer the results of the linear interpolation age model over the

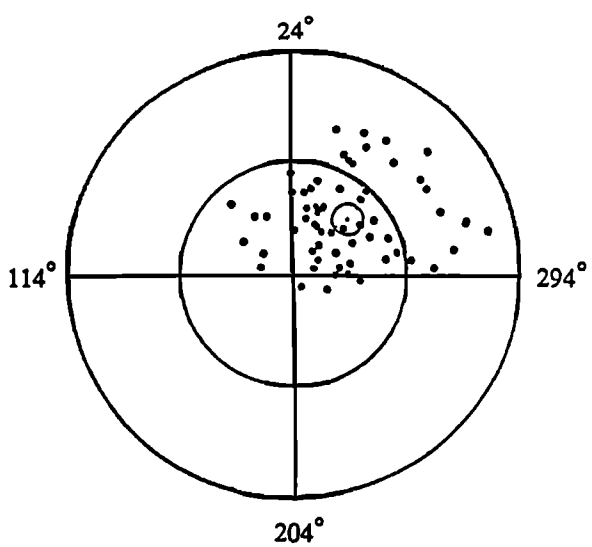


Fig. 15. Stereographic projections of VGPs. Each VGP is calculated from the averaged inclination and declination over a 200-year window. The site has been rotated to the bottom of the graph in order to show the "far-sided and right-handed" distributions of the VGPs. The small circle inside the plot is the α95 circle of the mean VGP.

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TABLE 4. Angular Dispersion Calculation

Window Size, years	N	With Respect to Mean VGP			With Respect to Geographic Pole		
		S_l	S	S_u	S_l	S	S_u
100	112	6.10	6.67	7.40	8.77	9.59	10.63
250	50	5.26	5.99	6.97	8.11	9.23	10.74
500	26	4.25	5.08	6.31	7.15	8.55	10.62

The calculations are done relative to the mean VGP and the geographic pole, respectively, after the data were thinned using the three different windows. Averages are $5.9^{+1.0}_{-0.7}$ with respect to mean VGP and $9.1^{+1.5}_{-1.1}$ with respect to geographic pole. N is number of VGPs averaged, S is angular dispersion, and S_l, S_u are lower and upper limits on S at 95% level.

linear regression age model and use the former model in subsequent spectral analyses.

Comparison With Lava Flow Data

Comparisons of the inclination and declination records from Lake Waiau sediments and Hawaiian lava flows are shown in Figure 20.

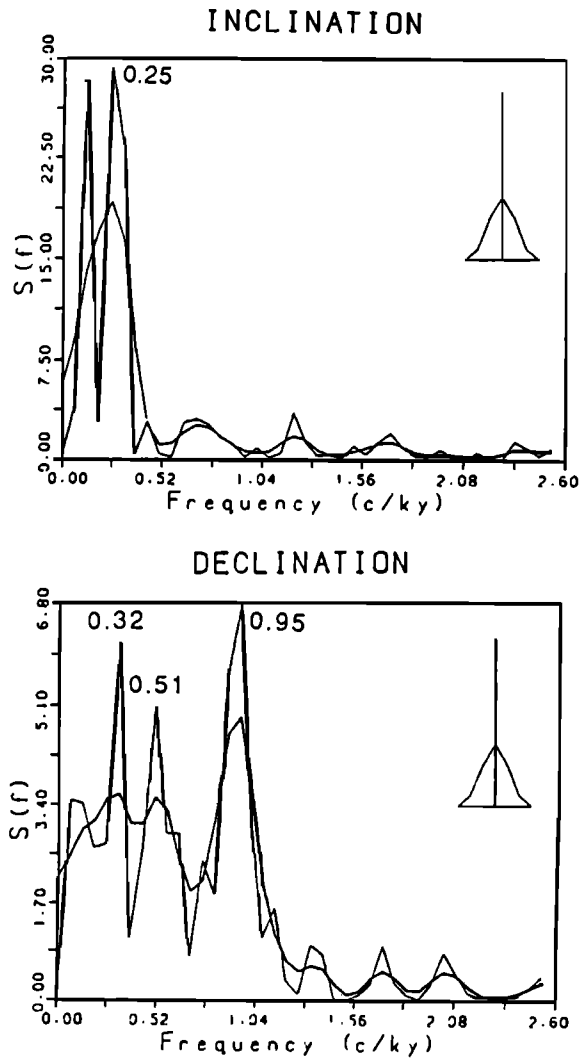


Fig. 16. Spectrums of inclination and declination time series analyses of core 82-2. Both unsmoothed and smoothed periodograms are shown. The Nyquist frequency is 8.06 cycles/kyr.

Paleomagnetic data from the lake sediments and the lava flows have similar patterns and amplitudes. An offset in the age of the first major inclination trough between the lake sediment and lava flow data is seen at approximately 2500 years B.P. This discrepancy can be caused either by an error in ^{14}C dating of the lava flows, the small number of ^{14}C dates from the lake sediments, or a possible lag effect produced by the postdepositional detrital remnant magnetization (DRM) processes in the lake sediments. We are assuming that DRM is postdepositional because the water content of the upper few cm of the sediments is very high and the directional data from these sediments was very scattered.

Both the lake sediments and lava flows have a mean inclination of about 27°, which is significantly lower than the 35° predicted by the geocentric axial dipole hypothesis. The mean declination is about 5° which is higher than the 0° predicted by the geocentric axial dipole hypothesis. It is well known that inclination errors in sediments can be caused by some sedimentologic processes, such as compaction [*Verosub*, 1977]. Bioturbation rates are low in Lake Waiau evidenced by the preservation of fine-scale laminations. Therefore, bioturbation is neither a viable mechanism for smoothing of the SV record, nor production of an inclination error. However, shallow inclinations are also a characteristic of Hawaiian

TABLE 5. Frequencies and Their Equivalent Periodicities of the Major Inclination and Declination Peaks Obtained Using the Periodogram Method and MEM With Different Choices of Parameter *m*

Inclination	Declination
Frequencies, cycles/kyr	
$0.13^{+0.00}_{-0.00}$	$0.13^{+0.03}_{-0.01}$
$0.28^{+0.01}_{-0.02}$	$0.31^{+0.01}_{-0.05}$
	$0.50^{+0.02}_{-0.02}$
	$0.94^{+0.03}_{-0.01}$
Periodicities, cycles/kyr	
$7.86^{+0.05}_{-0.05}$	$7.84^{+1.01}_{-1.51}$
$3.63^{+0.32}_{-0.23}$	$3.27^{+0.14}_{-0.44}$
	$2.01^{+0.10}_{-0.06}$
	$1.07^{+0.03}_{-0.01}$

The average values are shown above with the maximum value in superscript and minimum value in subscript.

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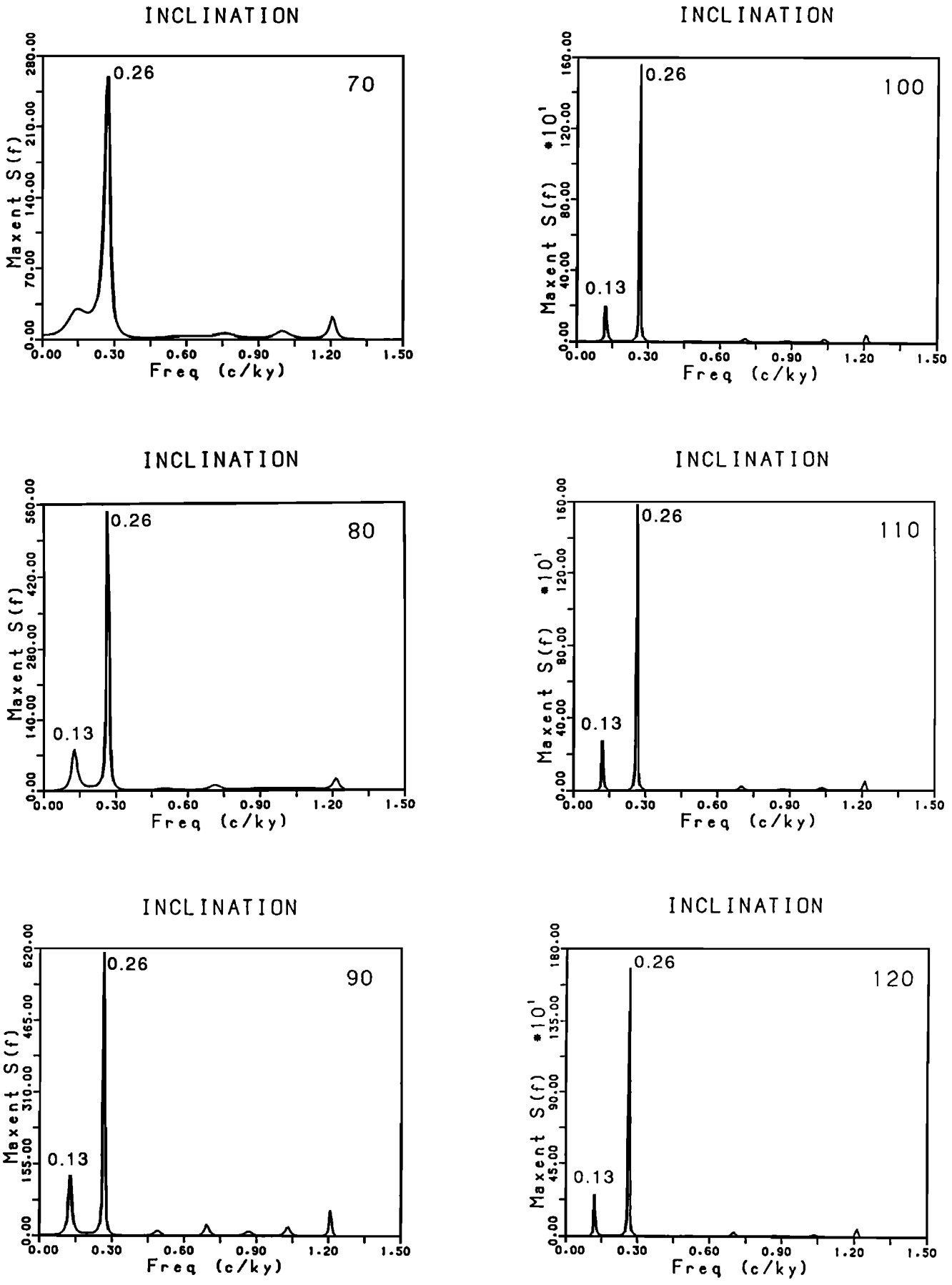
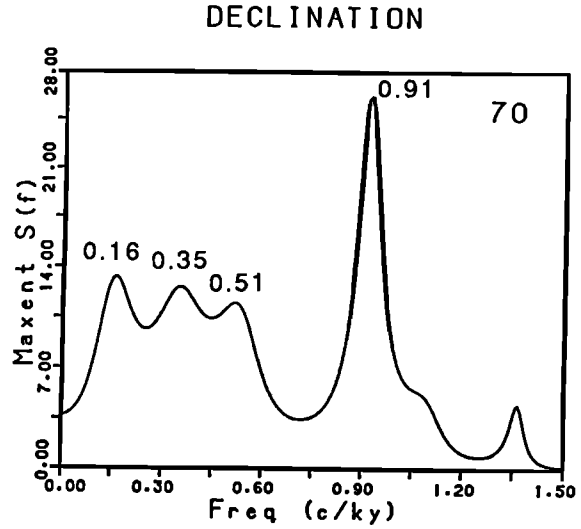
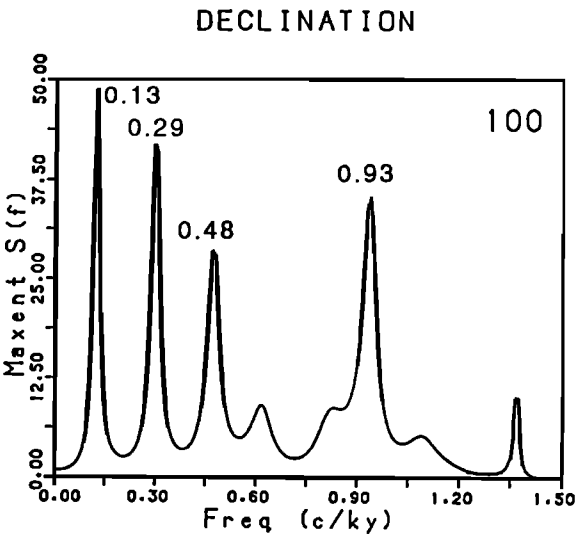
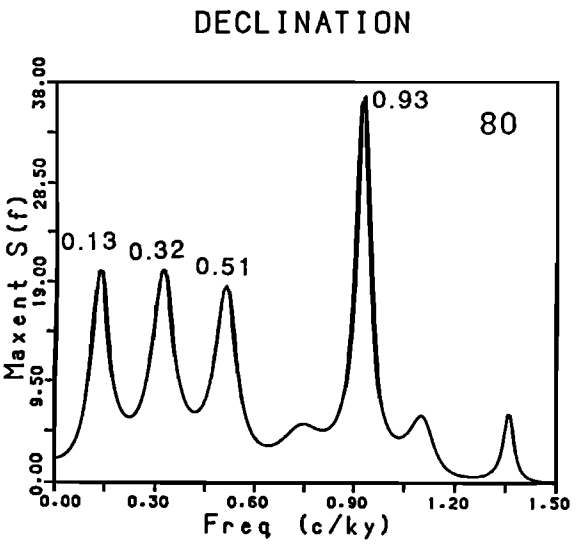


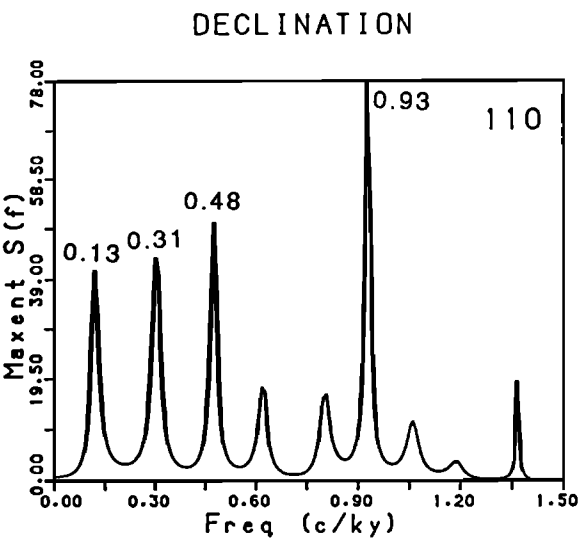
Fig. 17. Inclination and declination spectrums obtained from MEM with different choices of parameter *m*.

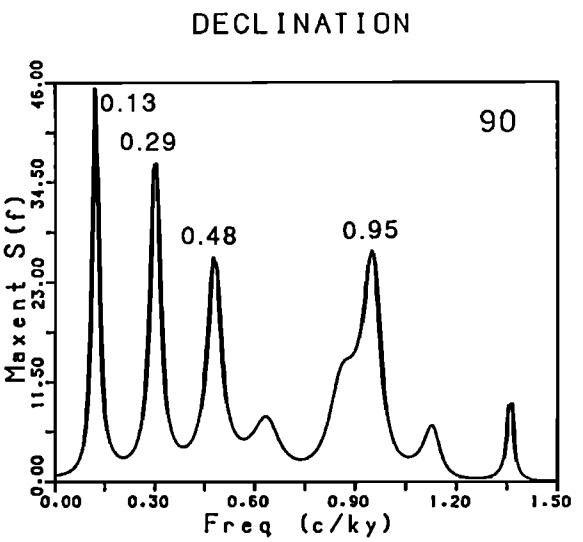
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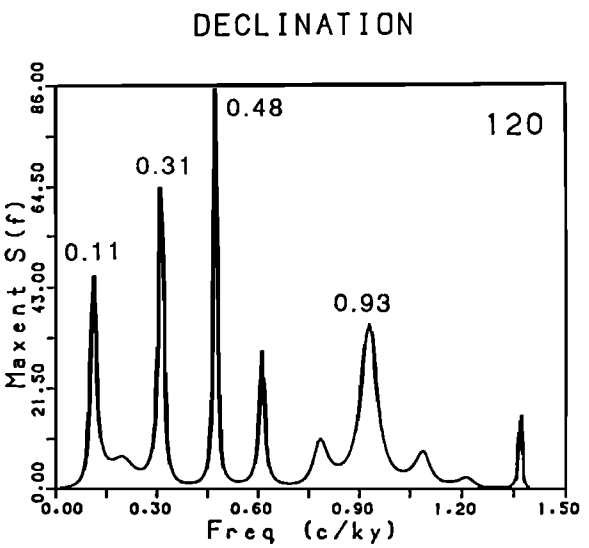


Fig. 17. (continued)

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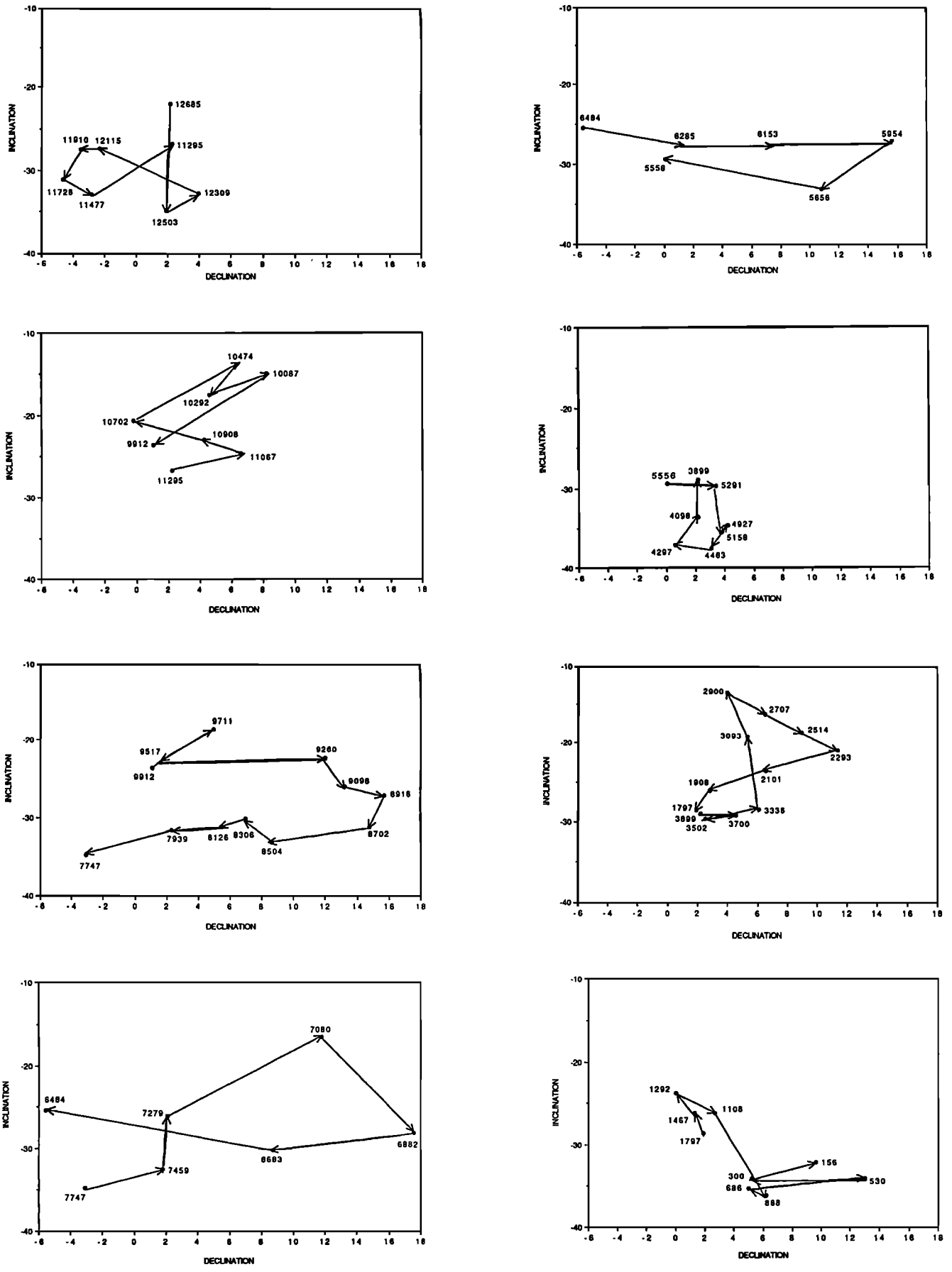


Fig. 18. The looping patterns of the geomagnetic field vectors. The numbers beside each dot are radiocarbon age in years B.P.

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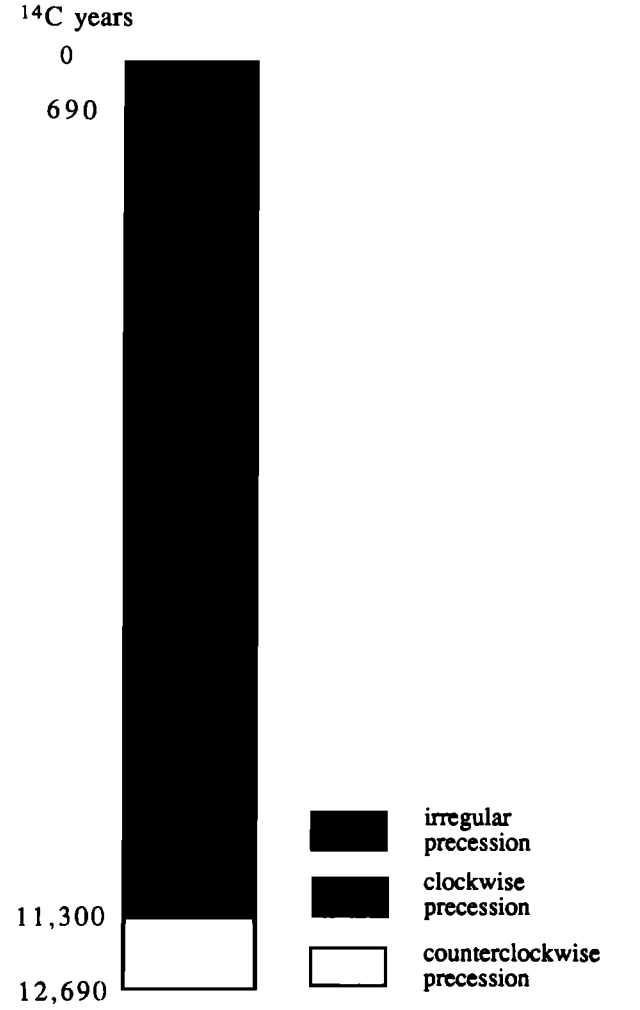


Fig. 19. The generalized diagram of precession patterns of the geomagnetic field vectors.

basalts for at least the past five million years [*Castro and Brown*, 1987]. *Castro and Brown* [1987] attributed shallow inclinations to the inaccurate recording of the geomagnetic field by the lava flows. Although we cannot completely rule out this possibility, the striking similarity between the lake sediment and the lava flow data suggests that the geomagnetic field rather than the recording mechanism is the cause of the shallow inclinations.

VGPs transformed from the observed inclination and declination values show a "far-sided, right-handed" tendency in both the lake sediments and the lava flows [*McWilliams et al.*, 1982]. Overall, the lake sediment and lava flow data sets are very similar with respect to the amplitude and age of the major features, the shallow average inclination values, and the VGP distribution.

VGP Angular Dispersion and the Pacific Nondipole Low

The discrepancy between the calculated angular dispersion values from the lake sediment data (this study) and the lava flow data [*McWilliams et al.*, 1982] is probably due to the nonuniform temporal distribution of the lava flow data. Although the lava flow data are thinned using different windows, the thinned data are not uniformly distributed (Figure 21). The data are much more densely distributed for the last 6000 years than for the interval from 6000 to 15,000 years B.P. The amplitude of variation in the inclination and the declination for consecutive data points is greater for the intervals with sparsely distributed data than for the intervals with densely distributed data (Figure 21), resulting in greater angular dispersion from sparse data intervals. This type of sampling bias in the lava flow studies might explain the smaller angular dispersion calculated from the densely distributed lake sediment data.

The low angular dispersion for Hawaii calculated from both lake sediments and lava flows could be due either to low SV, to incomplete averaging of SV, or both. The time scale for PSV is usually 10^4 years or more, because the rapidly varying part of nondipole field changes over periods up to about 10^3 years, whereas the dipole field changes over periods of about 10^4 years [*Cox and Doell*, 1964]. Paleomagnetic studies of sediments from two lakes in Minnesota by *Lund and Banerjee* [1985] suggest that the time-averaged field over 10^4 years is an axial geocentric dipole. The above considerations suggest that it is more likely that the regional geomagnetic field in the central North Pacific rather than an incomplete averaging of PSV produced the anomalously low SV and it has been a persistent geomagnetic feature during the last 13,000 years.

The low SV in the central Pacific can be explained by regional irregularities in temperature, topography or electrical conductivity at the core-mantle boundary (CMB). *Vestine and Kahle* [1966] proposed that if there is a downflow of fluid in the core, the SV may then be reduced in intensity relative to those regions experiencing an upflow. Downwelling flow in the core can be produced by a cold spot in the lower mantle and a depression of the CMB [*Bloxham and Gubbins*, 1985, 1987]. In addition, a highly conducting region in the mantle could screen part of the SV at the Earth's surface [*Bloxham et al.*, 1989]. In either case, the low SV in the central Pacific should be a long-term feature of the geomagnetic field if it is caused by these mechanisms.

Magnetic Spectrum and Precession Pattern

The frequencies obtained from spectral analyses can be divided into two bands, the low-frequency band consisting of an approximately 8000-year periodicity and a high-frequency band consisting of approximately 1100-, 2000-, and 3500-year periodicities. The approximately 10^4-year fluctuation was referred to as dipole variability by *Barton* [1983] and *Lund and Banerjee* [1985]. The 3500-year periodicity is probably derived from the source of nondipole field variation. The approximately 1100- and 2000-year periodicities only occur in the declination record. We do not understand the exact nature of these periods, but they are probably derived either from geomagnetic sources that affect the declination only, or from the offsets between sections in the core, which will only affect declinations.

The precession pattern (Figure 19) shows no drift for the last 700 years, which is consistent with the observation by *Bloxham et al.* [1989] that no westward drift of the field has occurred in the central Pacific during the last few hundred years. Prior to the last 700 years, the SV records are dominated by clockwise precession. Clockwise precession has traditionally been attributed to westward drift of the geomagnetic field according to Runcorn's rule. Therefore, we infer that westward drift was probably a long-term characteristic of the geomagnetic field in Hawaii. However, the counterclockwise precession observed from approximately 12,700 years B.P. to 11,300 years B.P. implies the eastward drift may have predominated during that interval.

CONCLUSIONS

1. The data collected from radiocarbon-dated lake sediments in Hawaii are similar to the paleomagnetic data obtained from lava flows, but are of much higher resolution. The shallow average inclination recorded in the sediments and lava flows is probably a real feature of the regional geomagnetic field and is not a product of problems with recording mechanisms.

2. The angular dispersion calculated using the continuous

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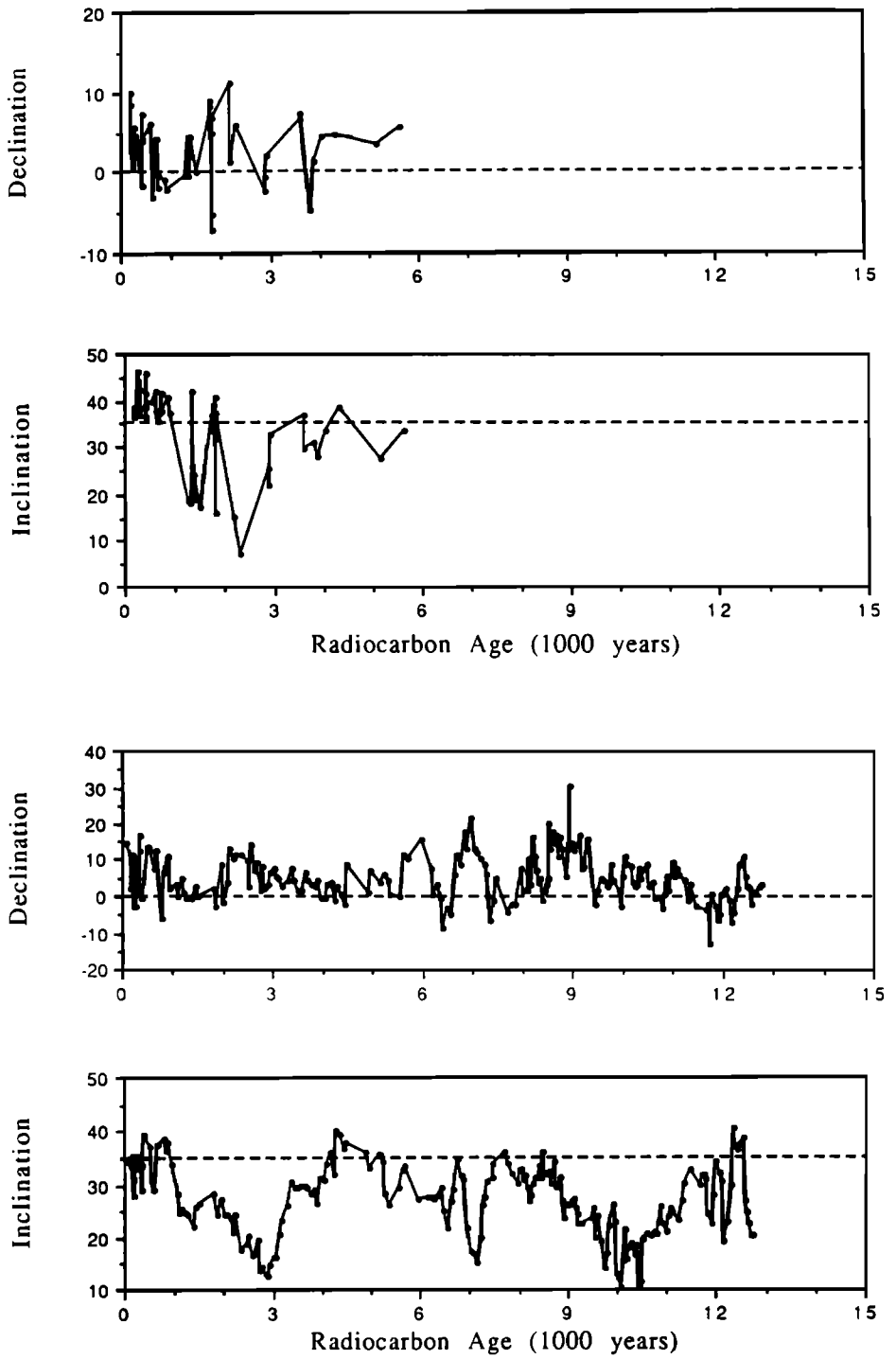


Fig. 20. Comparison of inclination and declination data from Lake Waiau sediments (bottom) and lava flows (top) [*Holcomb et al.*, 1986].

paleomagnetic data from Hawaii is lower than that predicted by the global PSV model G [*McFadden et al.*, 1988], supporting the hypothesis that anomalously low SV has occurred beneath the central Pacific region on time scales of 10^4 years. Regional irregularities in temperature, topography or electrical conductivity at the core-mantle boundary are inferred as a cause of the low SV.

3. The waveform of the PSV curve for Hawaii suggests a stable wave dynamo process. The approximately 8000- and 3500-year periodicities shown in both inclination and declination spectrums may represent dipole and nondipole variation, respectively. The approximately 1100- and 2000-year periodicities shown in the declination spectrum are probably derived from either the sources that affect declination only or the offsets between core sections.

4. There is no apparent drift of the geomagnetic field for the last 700 years. The SV is dominated by the clockwise looping of the geomagnetic vectors during the interval 700-11,300 years B.P. This type of behavior has traditionally been attributed to westward drift of the nondipole field. We infer that westward drift may have been a long-term feature of the geomagnetic field in Hawaii but that no apparent drift has occurred during the last few centuries.

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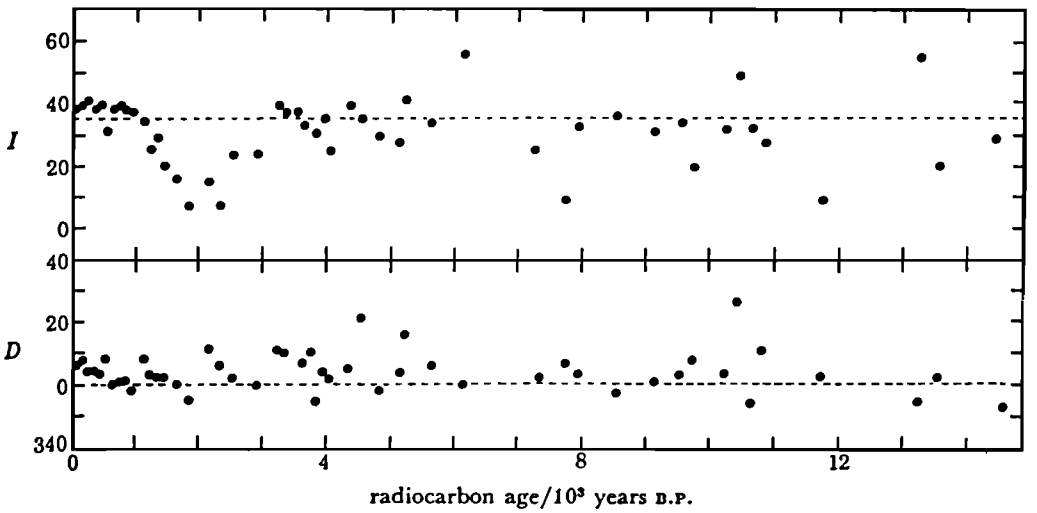


Fig. 21. Inclination and declination data from lava flows [*McWilliams et al.*, 1982].

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